

# Buffer solutions (HL)

## Higher level (HL)

### Learning outcomes

By the end of this section you should be able to:

- (R3.1.16) Describe the composition of acidic and basic buffers and explain their actions.
- (R3.1.17) Solve problems involving the composition and pH of a buffer solution, using the equilibrium constant.

We have seen solutions that are acidic, alkaline and neutral and the effect on pH when these solutions undergo titration. There are many natural systems at equilibrium where the pH of the solution needs to remain constant to protect the organism or ecosystem.

One example is your blood. The pH of your blood remains fairly constant, even when acidic or alkaline substances travel through your bloodstream. If the pH in your blood changes, it would have an effect on your health, leading to respiratory difficulties, liver failure, even brain damage.

When a solution resists changes in pH when acidic or alkaline substances are added, it is known as a buffer solution or simply a buffer. Buffer solutions have many uses in chemistry – they are utilised whenever it is necessary to have a controlled pH. Buffers are used to calibrate pH meter and are also an important ingredient in many consumer products, such as shampoos and dyes.

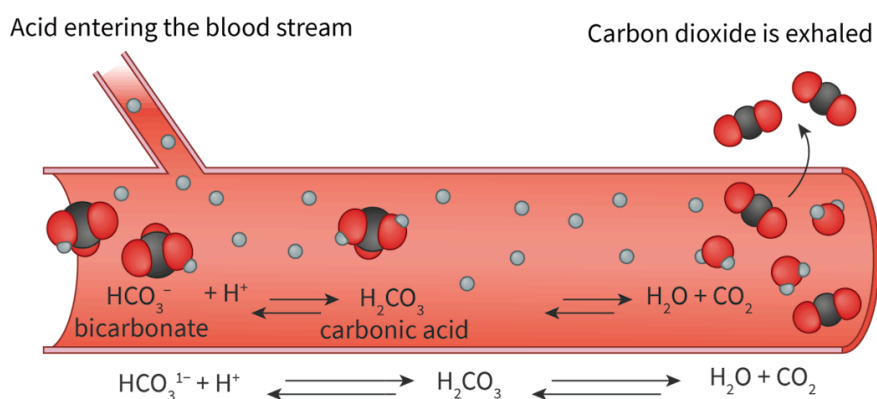


Figure 1. The buffer in blood.

More information for figure 1

The image is a diagram illustrating the buffer system in blood. It depicts a blood vessel with molecules and chemical reactions labeled within it. On the left side, labeled "Acid entering the blood stream," a reaction is shown between hydrogen ions ( $\text{H}^+$ ) and bicarbonate ions ( $\text{HCO}_3^-$ ) to form carbonic acid ( $\text{H}_2\text{CO}_3$ ). This is further detailed with an arrow pointing left showing the equation  $\text{HCO}_3^- + \text{H}^+ \leftrightarrow \text{H}_2\text{CO}_3$ . To the right, the conversion of carbonic acid

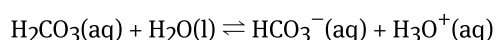
to carbon dioxide (CO<sub>2</sub>) and water (H<sub>2</sub>O) is illustrated with arrows and the equation  $\text{H}_2\text{CO}_3 \leftrightarrow \text{H}_2\text{O} + \text{CO}_2$ . On the far right, labeled "Carbon dioxide is exhaled," CO<sub>2</sub> molecules are depicted exiting the blood vessel. These interactions demonstrate how blood buffers changes in pH via reversible reactions.

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## Buffer solutions

How exactly does a buffer solution work to resist a change in pH? Recall that a system at equilibrium that undergoes a stress, such as the addition of small amounts of acid or base, will respond by shifting the position of equilibrium (see [section R2.3.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/)).

The buffer system in your blood uses carbonic acid and carbonate ions to maintain the pH of blood at an average 7.40.



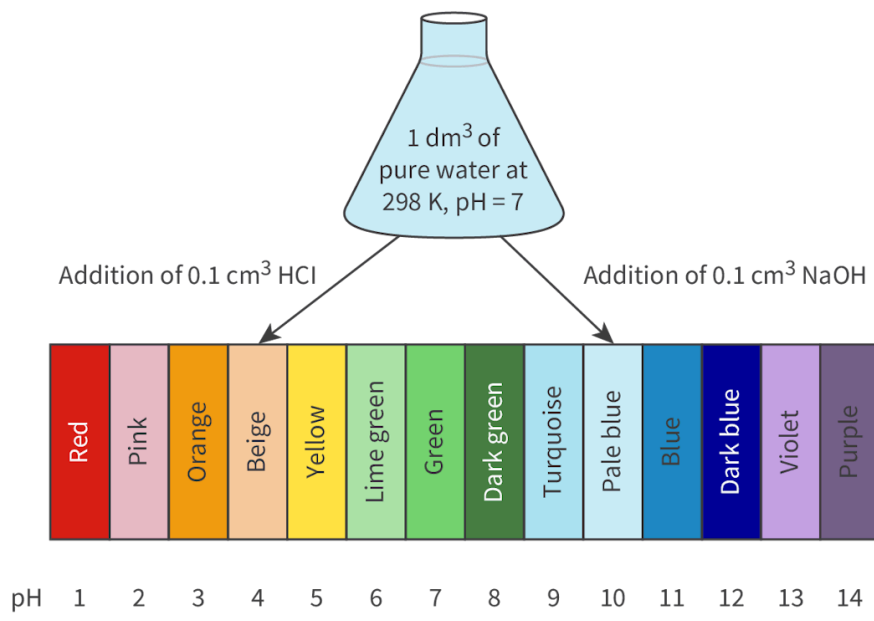
When a small amount of acid is added to a solution, the pH would be expected to decrease due to the addition of H<sup>+</sup> ions to the solution. When small amounts of acid are added to a system at equilibrium, the system can respond by shifting the position of equilibrium to minimise the stress. According to Le Chatelier's principle, the increase in concentration of the H<sup>+</sup> ions introduced by the acid, will cause the position of equilibrium to shift to the left (to favour the reactants), consuming the added H<sup>+</sup> ions and therefore restoring the pH to 7.40.



When a small amount of base is added to a solution, the pH would be expected to increase due to the addition of OH<sup>-</sup> ions to the solution. When small amounts of base are added a buffer, the system can respond by shifting the position of equilibrium. According to Le Chatelier's principle, the increase in concentration of the OH<sup>-</sup> ions introduced by the base, will react with the H<sup>+</sup> ions in solution, causing the concentration of H<sup>+</sup> to decrease. This will cause the position of equilibrium to shift to the right (to favour the products), replacing the lost H<sup>+</sup> ions and therefore restoring the pH to 7.40.



Figure 2 shows the effect of adding a small amount of acid or alkali to pure water. Note the significant change in pH that occurs from pH 7 to either pH 4 (for the addition of an acid) or to pH 10 (for the addition of an alkali). If the same amount of acid or base were added to a buffer solution, we could expect to see a much smaller change in the pH.



**Figure 2.** The change in pH on addition of a small amount of acid or alkali to pure water.

 More information for figure 2

The diagram illustrates the change in pH of water when a small amount of acid (HCl) or alkali (NaOH) is added. At the center of the image is a conical flask labeled '1 dm<sup>3</sup> of pure water at 298 K, pH = 7'. Two arrows extend from this flask: one pointing left, labeled 'Addition of 0.1 cm<sup>3</sup> HCl', and the other pointing right, labeled 'Addition of 0.1 cm<sup>3</sup> NaOH'. Below the arrows, a rectangular color scale displays pH levels with corresponding colors, ranging from pH 1 to 14. The colors range from red (pH 1) progressing through pink, orange, beige, yellow, lime green, green, dark green, turquoise, pale blue, blue, dark blue, violet, to purple (pH 14). The left arrow (addition of HCl) indicates a decrease in pH towards 4 (beige), while the right arrow (addition of NaOH) indicates an increase towards pH 10 (pale blue).

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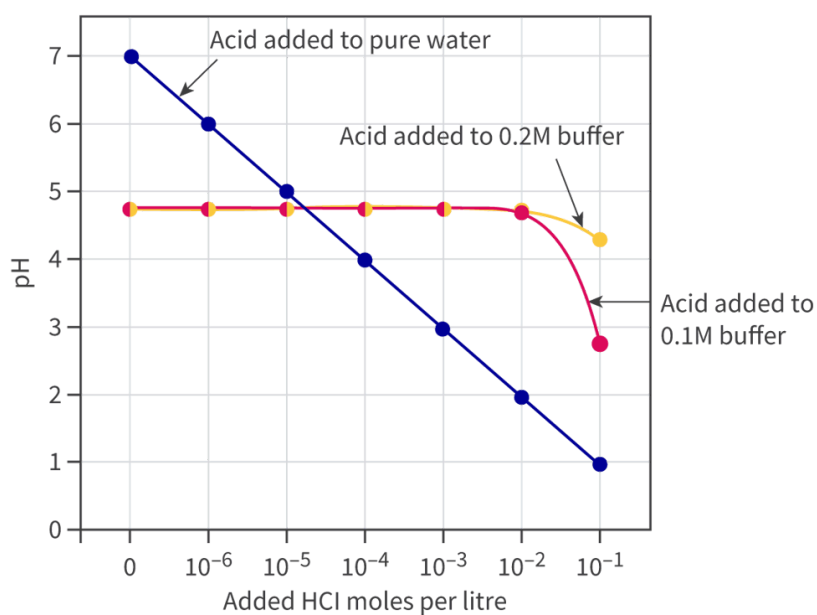
Watch **Video 1** to see what would happen if an acid or base was added to a buffer solution.

## Buffer Demonstration 2 0 for Avid



### Video 1. How a buffer keeps pH constant until a certain point.

Adding small amounts of acid or base to a buffer solution maintains the pH relatively constant. This comparative effect on pH is shown in **Figure 3**.



**Figure 3.** The change in pH on addition of a small amount of acid or alkali to pure water and to a buffer solution.

More information for figure 3

The graph illustrates how the pH changes when small amounts of hydrochloric acid (HCl) are added to pure water and two types of buffer solutions. The X-axis of the graph is labeled "Added HCl moles per litre" and presents a logarithmic scale from 0 to  $10^{-1}$ . The Y-axis is labeled "pH" and ranges from 0 to 7.

Three different curves are shown on the graph: 1. A blue line showing the acid added to pure water. It starts at the upper left corner with a pH of 7 and decreases steadily as more acid is added. 2. A yellow line indicating the data for a 0.2M buffer solution. The pH remains constant around 5 even with an increase in HCl up to  $10^{-3}$  moles per litre, then drops slightly. 3. A red line

showing the 0.1M buffer solution. Initially, the pH remains around 5 but begins to decline gradually after  $10^{-3}$  moles per litre, dropping significantly past  $10^{-2}$  moles per litre.

Overall, the graph demonstrates the buffering capacity, showing that adding acid to buffer solutions maintains a relatively constant pH compared to pure water.

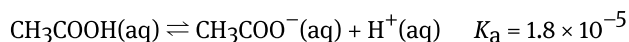
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## Theory of Knowledge

Buffer solutions were originally used to simulate biological systems. Experiments involving live cell cultures typically require a buffer medium to maintain pH for optimal biological processes. How does the concept of equilibrium in buffer solutions relate to the nature of scientific knowledge?

## Composition of buffers

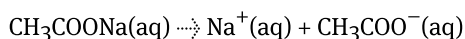
Recall that weak acids and bases ionise partially, existing as a system at equilibrium, whereas strong acids and bases ionise completely (see [section R3.1.6a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>)). Therefore, buffers require a weak acid or base as the primary component. For example, ethanoic acid is a weak acid that can be used to make a buffer solution. The dissociation of ethanoic acid and its  $K_a$  value are shown here.



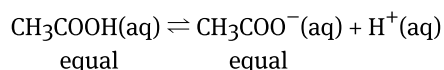
In order to make a buffer solution that can respond to stress, the system at equilibrium must contain a relatively equal amount of both the weak acid and the conjugate base ( $\text{CH}_3\text{COOH} / \text{CH}_3\text{COO}^-$ ) – otherwise, the response to the stress would not be very effective.

However, a weak acid or base does not ionise enough to provide a sufficient concentration of the conjugate base, since its  $K_a$  is very low. Therefore, the conjugate base must be added to the weak acid to provide a sufficient concentration.

How can the conjugate base ( $\text{CH}_3\text{COO}^-$ ) be added to the solution? Recall that salts can have acid–base activity (see [section R3.1.12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-of-salt-solutions-hl-id-46060/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-of-salt-solutions-hl-id-46060/>)). The ethanoate ion can be added to the solution by adding a salt that contains the ethanoate ion as the anion, but care must be taken to select a cation that will not provide any acid–base activity to the buffer solution. An example of a suitable salt is sodium ethanoate,  $\text{CH}_3\text{COONa}$ , which dissociates according to the following equation:

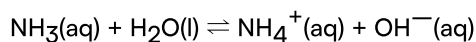


The combination of equal amounts of ethanoic acid and sodium ethanoate will provide relatively equal amounts of the weak acid and its conjugate base at equilibrium, allowing for effective buffer action.



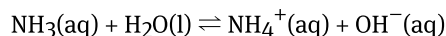
Using your knowledge of how to make a buffer solution starting with a weak acid, how would you make a buffer starting with a weak base, such as ammonia,  $\text{NH}_3$ ?

The weak base ammonia,  $\text{NH}_3$ , must also have its conjugate acid present in a relatively equal concentration for the buffer to respond effectively to stress.



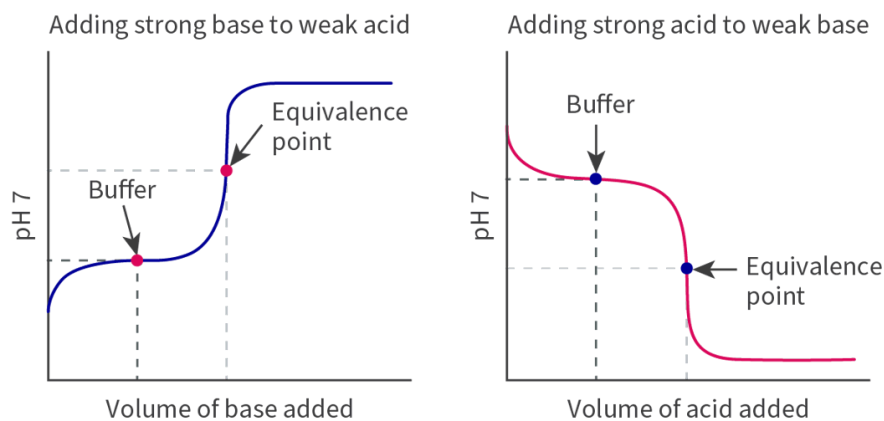
The conjugate acid of ammonia is ammonium,  $\text{NH}_4^+$ . Ammonium ions can be added by using a salt that contains ammonium ions as the cation, but the anion must not have any acid–base activity. One example for a suitable salt is ammonium chloride,  $\text{NH}_4\text{Cl}$ .

For this buffer system, the equilibrium equation is:



If small amounts of acid are added, the  $\text{H}^+$  ions from the acid will neutralise the  $\text{OH}^-$  ions in solution, decreasing their concentration and causing the equilibrium to shift to the right to replace them and restore the pH. Conversely, if small amounts of base are added, the concentration of  $\text{OH}^-$  ions in solution increases, causing the equilibrium to shift to the left to consume them, which then restores the pH.

There is another way to make a buffer solution. Can you recall another time that you have seen the term buffer before. In [section R3.1.13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-of-strong-and-weak-monoprotic-acids-and-bases-hl-id-46061/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-of-strong-and-weak-monoprotic-acids-and-bases-hl-id-46061/>) a weak acid (or base) was titrated with a strong base (or acid) before reaching the equivalence point. There is an area on the titration curve known as the buffer region. This is where the partially titrated acid(or base) contains a sufficient amount of its conjugate base (or acid), such that a buffer solution is made (**Figure 4**).



**Figure 4.** A buffer can be made by partially neutralising a weak acid or base.

 More information for figure 4

The image contains two graphs illustrating titration curves.

On the left, the graph titled "Adding strong base to weak acid" displays a blue titration curve. The x-axis is labeled "Volume of base added," and the y-axis is labeled "pH." The curve starts at a low pH, increases gradually, then sharply rises, and finally levels off after the equivalence point. The buffer region is marked and identified before the steep rise begins. The equivalence point is indicated.

On the right, the graph titled "Adding strong acid to weak base" shows a red titration curve. The x-axis is labeled "Volume of acid added," and the y-axis is again labeled "pH." This curve starts at a high pH, decreases gradually, then sharply drops, and eventually levels off beyond the equivalence point. The buffer region is marked and identified prior to the sharp decline. The equivalence point is also indicated in this graph.

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## Worked example 1

Which of the following mixtures forms a buffer solution with a pH below 7?

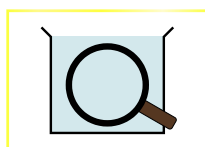
1.  $0.0200 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  carbonic acid is mixed with  $0.0100 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  sodium hydroxide.
2.  $0.0200 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  carbonic acid is mixed with  $0.0200 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  sodium hydroxide.
3.  $0.0100 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  carbonic acid is mixed with  $0.0200 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  sodium hydroxide.

1. will result in a buffer solution with a pH of less than 7 since the moles of strong base are half of the number of moles of weak acid, resulting in partially neutralising the weak acid.
2. will result in the acid is being completely neutralised since the moles of strong base are equal to the moles of weak acid. This solution will not act as a buffer solution.
3. will result in an excess of strong base as the moles of strong base are greater than the moles of weak acid. The pH is more than 7 as too much base is added. This solution will not act as a buffer solution.

These combinations show the mixing of a strong base and a weak acid. For a buffer to be formed, we saw that enough of the strong base should be added to half neutralise the weak acid. The only combination that matches this is 1.

## Study skills

If you are having a hard time visualising these processes, you can access the simulation below and see this at a molecular level:



## Interactive 1. Explore pH: Interactive Acid-Base Solutions Simulation.

 More information for interactive 1

An interactive simulation demonstrates acid-base solutions through two panels labeled Intro and My Solution, each of which zooms in on its corresponding illustration when selected.

In the intro panel, a panel on the right allows users to select a solution from the following options:

- Water ( $\text{H}_2\text{O}$ )
- Strong Acid (HA)
- Weak Acid (HA)
- Strong Base (MOH)
- Weak Base (B)

A beaker in the center contains 1 liter of a selected solution, with a magnifying glass revealing the particles inside. Users can immerse a pH meter into the solution to find its respective pH values.

Another panel on the right allows users to select different viewing options, such as displaying particles, a bar graph for equilibrium concentration, and hide views.

Below the beaker is the reaction of the respective solution. There are also options to switch the pH meter to a pH scale and a circuit for checking electrical conductivity.

The reaction and equilibrium concentrations of the reactants and products are as follows:

### For water:

- Reaction:  $2 \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$
- Equilibrium concentrations of  $\text{H}_2\text{O}$ ,  $\text{H}_3\text{O}^+$ , and  $\text{OH}^-$  are 55.6,  $1.0 \times 10^{-7}$ , and  $1.0 \times 10^{-7}$  moles per liter, respectively.
- The pH is 7.00.
- Poor conductor of electricity.

### For Strong Acid (HA):

- Reaction:  $\text{HA} + \text{H}_2\text{O} \rightarrow \text{A}^- + \text{H}_3\text{O}^+$
- Equilibrium concentrations of HA,  $\text{H}_2\text{O}$ ,  $\text{A}^-$ , and  $\text{H}_3\text{O}^+$  are negligible, 55.6,  $1.0 \times 10^{-2}$ , and  $1.0 \times 10^{-2}$  moles per liter, respectively.
- The pH is 2.00.
- Strong conductor of electricity.

### For Weak Acid (HA):

- Reaction:  $\text{HA} + \text{H}_2\text{O} \rightleftharpoons \text{A}^- + \text{H}_3\text{O}^+$
- Equilibrium concentrations of HA,  $\text{H}_2\text{O}$ ,  $\text{A}^-$ , and  $\text{H}_3\text{O}^+$  are  $1.0 \times 10^{-2}$ , 55.6,  $3.2 \times 10^{-5}$ , and  $3.2 \times 10^{-5}$  moles per liter, respectively.
- The pH is 4.50.
- Weak conductor of electricity.



**For Strong Base (MOH):**

- Reaction:  $M^+OH^- \rightarrow M^+ + OH^-$
- Equilibrium concentrations of  $M^+OH^-$ ,  $M^+$ , and  $OH^-$  are negligible,  $1.0 \times 10^{-2}$ , and  $1.0 \times 10^{-2}$  moles per liter, respectively.
- The pH is 12.00.
- Strong conductor of electricity.

**For Weak Base (B):**

- Reaction:  $B + H_2O \rightleftharpoons BH^+ + OH^-$
- Equilibrium concentrations of B,  $H_2O$ ,  $BH^+$ , and  $OH^-$  are  $1.0 \times 10^{-2}$ , 55.6,  $3.2 \times 10^{-5}$ , and  $3.2 \times 10^{-5}$  moles per liter, respectively.
- The pH is 9.50.
- Strong conductor of electricity.

In the My Solution panel, the same phenomenon is exhibited. Users can check the pH of the weak and strong acids and bases by changing the initial concentration, ranging from 0.001 to 1 mole per liter. Users can also adjust a slider to change the solution strength from weaker to stronger.

This simulation allows users to understand that as the initial concentration increases from 0.001 to 1 mole per liter, the pH of the weak acid decreases, the pH of the weak base increases, the pH of the strong acid decreases, and the pH of the strong base increases.

**Table 1** shows the four different ways to make a buffer:

**Table 1.** Ways to make a buffer.

|          | Begin with: | Then add:                                      | Final pH of buffer solution |
|----------|-------------|------------------------------------------------|-----------------------------|
| Method 1 | Weak acid   | Salt containing the conjugate base             | pH < 7                      |
| Method 2 | Weak base   | Salt containing the conjugate acid             | pH > 7                      |
| Method 3 | Weak acid   | A strong base to half neutralise the weak acid | pH < 7                      |
| Method 4 | Weak base   | A strong acid to half neutralise the weak base | pH > 7                      |

## Making connections

Can you think about the role of buffers and explain why buffer solutions must be composed of weak acid or base conjugate systems, not of strong acids or bases? What can weak acids/bases do that strong acids/bases cannot? (see [subtopic R2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

Because a buffer solution consists of an equilibrium system between acid and conjugate base, when adding a small amount of strong acid or base, Le Châtelier's principle can be applied to explain how the pH remains constant. How does Le Châtelier's principle enable us to interpret the behaviour of indicators and buffer solutions? (see [subtopic R2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

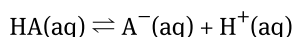
## The pH of a buffer

When preparing to make a buffer solution, it is important to select the appropriate components so that the buffer will have the desired pH required for the experiment. The pH of a buffer is determined by:

- the  $pK_a$  or  $pK_b$  of the weak acid (or base);
- the relative ratio of the concentrations of weak acid (or weak base) and the conjugate base (or conjugate acid)

## Calculating the pH of an acidic buffer

We saw that an acidic buffer solution consists of a weak acid HA and its conjugate base  $A^-$  in similar mole quantities. The concentration of the weak acid is represented as HA and the concentration of the conjugate base is represented as  $A^-$ . When the two solutions are added together, there will be a dilution that occurs and must be corrected.



Using the equilibrium expression and  $K_a$  (or  $pK_a$ ) of the weak acid,

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

When the  $[H^+]$  is determined, the pH is easily calculated using the formula,

$$pH = -\log[H^+]$$

## Worked example 2

A student is making a sample of simulated gastric juice in the stomach by preparing a buffer for an experiment.

$0.0200 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  carbonic acid ( $H_2CO_3$ ) is added to  $0.0100 \text{ dm}^3$  of  $0.10 \text{ mol dm}^{-3}$  sodium hydrogen carbonate ( $NaHCO_3$ ).

Calculate the pH of the buffer solution.

$K_a$  of carbonic acid is  $4.47 \times 10^{-7}$ .

First, the new concentrations must be calculated for dilution in the new total volume:

$$0.0200 \text{ dm}^3 + 0.0100 \text{ dm}^3 = 0.0300 \text{ dm}^3 \text{ total volume}$$

The concentration of carbonic acid is:

$$\frac{0.0200 \times 0.10}{0.0300} = 0.0667 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3$$

The concentration of sodium hydrogen carbonate is:

$$\frac{0.0100 \times 0.10}{0.0300} = 0.0333 \text{ mol dm}^{-3} \text{ NaHCO}_3$$

Since  $\text{NaHCO}_3$  dissolves in the aqueous solution, it is used as the initial concentration of the conjugate base before the equilibrium is reached.

Using an ICE table, fill in the information that is known. The initial concentrations of  $\text{H}_2\text{CO}_3$  and  $\text{NaHCO}_3$  were calculated and before the acid dissociates,  $[\text{H}^+] = 0$ .

The reaction then shifts to the right side to favour its formation of the products until equilibrium is established.

| Step/Variable                                                   | Equilibrium E                      |
|-----------------------------------------------------------------|------------------------------------|
| Let $x$ be the amount of acid dissociated to reach equilibrium. | $\text{H}_2\text{CO}_3(\text{aq})$ |
| Initial concentration ( $\text{mol dm}^{-3}$ )                  | 0.0667                             |
| Change in concentration ( $\text{mol dm}^{-3}$ )                | $-x$                               |
| Equilibrium concentration ( $\text{mol dm}^{-3}$ )              | $0.0667 - x$                       |

Substituting into the  $K_a$  equation gives:

$$\begin{aligned}
 K_a &= \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} \\
 &= \frac{x(0.0333 + x)}{0.0667 - x}
 \end{aligned}$$

Since the  $K_a$  is very small, and the acid is weak, it dissociates very little, and therefore  $x$  is a very small number and is negligible when being added or subtracted from a much larger number. Therefore:

$$\begin{aligned}
 0.0333 + x &= 0.0333 \text{ and} \\
 0.0667 - x &= 0.0667
 \end{aligned}$$

Solving for  $x$  gives:

$$\begin{aligned}
 x &= \frac{0.0667}{0.0333} \times 4.47 \times 10^{-7} \\
 &= 8.95 \times 10^{-7}
 \end{aligned}$$

Therefore:

$$[\text{H}^+] = 8.95 \times 10^{-7} \text{ mol dm}^{-3}$$

$$\begin{aligned} \text{pH} &= -\log[\text{H}^+] \\ &= -\log 8.95 \times 10^{-7} \\ &= 6.05 \end{aligned}$$

The pH of this buffer solution is reasonable, given that it is made using a weak acid. The pH would be expected to be below 7, but not close to 1. Always consider the result of your calculations to see that the pH value makes sense.

Also note the significant figures. Since the concentration of carbonic acid is given to 2 significant figures, the final result will have 2 digits to the right of the decimal place (for pH values the digit to the left of the decimal place has no effect on the number of significant figures).

## Calculating the pH of a basic buffer

Similarly, for a weak base B and its conjugate acid  $\text{HB}^+$ .



Therefore:

$$\begin{aligned} K_b &= \frac{[\text{OH}^-][\text{HB}^+]}{[\text{B}]} \\ -\log_{10} K_b &= -\log_{10} \left( \frac{[\text{OH}^-][\text{HB}^+]}{[\text{B}]} \right) \\ -\log_{10} K_b &= -\log_{10} [\text{OH}^-] - \log_{10} \left( \frac{[\text{HB}^+]}{[\text{B}]} \right) \\ \text{p}K_b &= \text{pOH} - \log_{10} \left( \frac{[\text{HB}^+]}{[\text{B}]} \right) \end{aligned}$$

Rearrange to find pOH:

$$\begin{aligned} \text{pOH} &= \text{p}K_b + \log_{10} \left( \frac{[\text{HB}^+]}{[\text{B}]} \right) \\ \text{pH} &= 14 - \text{pOH} \end{aligned}$$

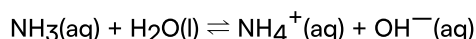
B is a weak base and does not dissociate completely and the initial concentration of the base B is the approximately the same as the final concentration and concentration of the conjugated acid  $\text{HB}^+$  stays approximately the same hence, we can use the initial concentrations.

### Worked example 3

Calculate the pH of the buffer solution prepared by combining 0.0667 mol ammonia and 0.0333 mol ammonium ions in a 1.00 dm<sup>3</sup> volumetric flask.

$K_b$  of ammonia is  $1.78 \times 10^{-5}$

The buffer system at equilibrium is:



Using an ICE table, fill in the information that is known. Since the total volume of the solution is  $1.00 \text{ dm}^3$ , the number of moles is equal to the molar concentration. Also note that the initial concentration of  $\text{OH}^-$  will be 0 since the weak base has not dissociated yet. Equilibrium will then shift to the right, favouring the formation of the products.

| Step/Variable                                                   | Equilibrium Equation     |                      |  |
|-----------------------------------------------------------------|--------------------------|----------------------|--|
| Let $x$ be the amount of base dissociated to reach equilibrium. | $\text{NH}_3(\text{aq})$ | $\rightleftharpoons$ |  |
| Initial concentration ( $\text{mol dm}^{-3}$ )                  | 0.0667                   |                      |  |
| Change in concentration ( $\text{mol dm}^{-3}$ )                | $-x$                     |                      |  |
| Equilibrium concentration ( $\text{mol dm}^{-3}$ )              | $0.0667 - x$             |                      |  |

Substituting in the  $K_b$  equation gives:

$$\begin{aligned}
 K_b &= \frac{[\text{OH}^-][\text{NH}_4^+]}{[\text{NH}_3]} \\
 &= \frac{x(0.0333 + x)}{0.0667 - x}
 \end{aligned}$$

Since the  $K_b$  is very small, and the base is weak, it dissociates very little, and therefore  $x$  is a very small number and is negligible when being added or subtracted from a much larger number. Therefore:

$$0.0333 + x = 0.0333 \quad \text{and} \quad 0.0667 - x = 0.0667$$

Solving for  $x$  gives:

$$\begin{aligned}
 x &= \frac{0.0667}{0.0333} \times 1.78 \times 10^{-5} \\
 &= 3.56 \times 10^{-5}
 \end{aligned}$$

Therefore:

$$\begin{aligned}
 [\text{OH}^-] &= 3.56 \times 10^{-5} \text{ mol dm}^{-3} \\
 \text{pOH} &= -\log[\text{OH}^-] \\
 \text{pOH} &= -\log[3.56 \times 10^{-5} \text{ mol dm}^{-3}] \\
 &= 4.448 \\
 \text{pH} &= 14 - \text{pOH} \\
 \text{pH} &= 14 - 4.448 \\
 &= 9.55 \text{ (3 s.f.)}
 \end{aligned}$$

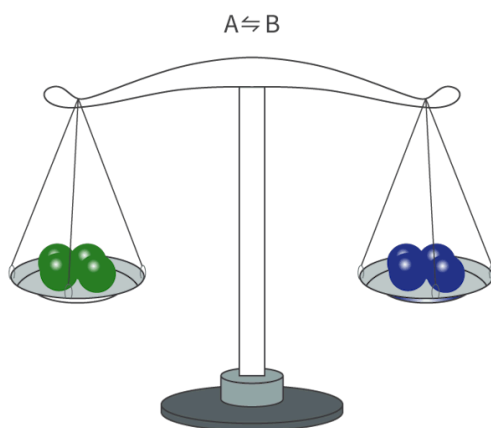
The pH of this buffer solution is therefore 9.552. This makes sense considering that the buffer is made with a weak base. The pH would be expected to be above 7, but not too close to 14. Always consider the result of your calculations to see that the pH value makes sense.

Also note the significant figures. Since the  $K_b$  is given to 3 significant figures, the final result of the logarithmic function in the pH calculation will have 3 digits to the right of the decimal place (for pH values the digit to the left of the decimal place has no effect on the number of significant figures).

## Dilution of a buffer solution

Would diluting a buffer solution with distilled water affect its pH? Think about the components of a buffer solution. When adding water, the ratios of the weak acid: conjugate base (or weak base: conjugate acid) will remain the same, but the concentrations will be lower, so the response of the buffer will be reduced.

Consider the analogy of a balance in **Figure 5**. When the buffer solution is an equilibrium, the two sides are balanced. If a stress is applied to the system by adding small quantities of acid or base, one pan will drop. Le Châtelier's principle allows for balls to be shifted from the heavy side to the lighter side, restoring equilibrium.



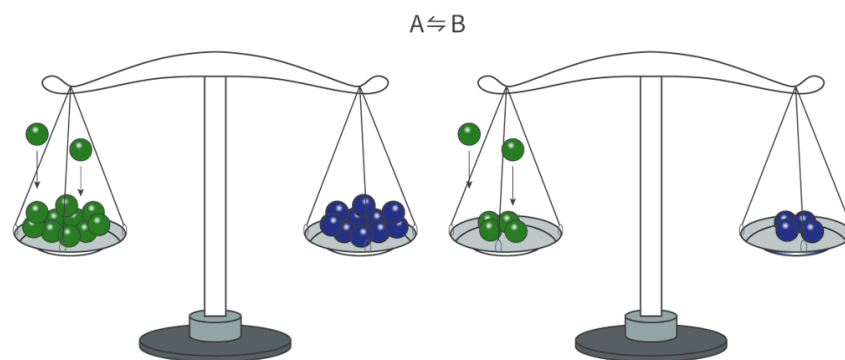
**Figure 5.** A buffer contains a balanced ratio of a weak acid (or base) and its conjugate base (or acid).

 More information for figure 5

The image is an illustration of a balance scale with two pans. On the left pan, there are two green spheres, while on the right pan, there are two blue spheres. Both sides are shown to be in equilibrium, representing a balanced state. This visual serves as a metaphor for a chemical buffer system, where the balance illustrates the equilibrium maintained between a weak acid and its conjugate base, or a weak base and its conjugate acid. The image conveys the concept that in a buffer system, similar to the balance scale, equilibrium can be maintained even when stress, in the form of a small amount of acid or base, is applied.

[Generated by AI]

Consider that there are two different balances. One balance has ten balls on each side and the other balance contains only four on each side. When a stress is applied, let's say two additional balls are added (**Figure 6**), which balance can respond to stress better?



**Figure 6.** A more concentrated buffer can respond better to stress than a diluted buffer.

 More information for figure 6

The illustration depicts two balance scales, each with two sets of spheres representing buffer components. The balance on the left contains a larger number of green spheres on each side, with an additional two green spheres being added to one side, showing an imbalance. The balance on the right has fewer blue spheres on both sides with two additional spheres added on one side, creating a similar imbalance. Above the image, there's an equation symbol ' $A \rightleftharpoons B$ ', which represents chemical equilibrium. The image illustrates the concept of balanced versus stressed buffer systems, where a more concentrated buffer can better absorb additional reagents without losing equilibrium, compared to a diluted buffer.

[Generated by AI]

Both solutions are still buffers and will still respond by shifting the position of equilibrium to minimise the stress applied but remember that buffers can only resist pH changes when small amounts of acid or base are added. For a more concentrated buffer, the small amount of acid or base can be easily managed in relation to the higher concentrations of the buffer components, allowing equilibrium to be restored and the pH maintained. For a diluted buffer, there is not a high enough concentration of the buffer components to shift equilibrium sufficiently. Equilibrium cannot be restored and the pH will not be maintained.

### Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 15 minutes
- **Activity type:** Pair activity

### Instructions

Go on a hunt!

1. Research four or five buffer systems found in nature. Then do some research and fill in the blanks with the corresponding information in

the table below.

| Buffer | $pK_a$ or $pK_b$ | Formula of the conjugate acid | F |
|--------|------------------|-------------------------------|---|
|        |                  |                               |   |
|        |                  |                               |   |
|        |                  |                               |   |
|        |                  |                               |   |

1. Design an experiment that you could perform in the lab to prepare a buffer of pH5 and test its buffer capacity. What should you start with? Which materials would you need? Which calculations are needed? Which method can you use to test the buffer capacity?